

```
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score
import matplotlib.pyplot as plt
import kagglehub
```

```
path = kagglehub.dataset_download("adelansez/taxi-trips-chicago-2024")
print(path)
```

Using Colab cache for faster access to the 'taxi-trips-chicago-2024' dataset.
/kaggle/input/taxi-trips-chicago-2024

```
import os
print(os.listdir(path))
```

```
['Taxi_Trips_-_2024_20240408.csv']
```

```
data = pd.read_csv(path + "/Taxi_Trips_-_2024_20240408.csv")
data = data[['Trip Miles', 'Trip Total']]
print(data.shape)
data.head()
```

```
(865247, 2)
```

	Trip Miles	Trip Total
0	0.09	38.75
1	3.00	12.00
2	5.84	26.70
3	13.36	47.10
4	6.13	22.82

```
data.dropna(inplace=True)
print(data.shape)
data.isnull().sum()
```

```
(863196, 2)
```

	0
Trip Miles	0
Trip Total	0

```
dtype: int64
```

```
def remove_outliers(df, column):
    Q1 = df[column].quantile(0.25)
    Q3 = df[column].quantile(0.75)
    IQR = Q3 - Q1
    lower = Q1 - 1.5 * IQR
    upper = Q3 + 1.5 * IQR
    return df[(df[column] >= lower) & (df[column] <= upper)]
```

```
data = remove_outliers(data, 'Trip Miles')
data = remove_outliers(data, 'Trip Total')
print(data.shape)
```

```
(846790, 2)
```

```
x = data.iloc[:, :1].values
y = data.iloc[:, 1:2].values
```

```
x_train, x_test, y_train, y_test = train_test_split(x, y, test_size=0.2, random_state=42)
```

```
model = LinearRegression()
model.fit(x_train, y_train)
```

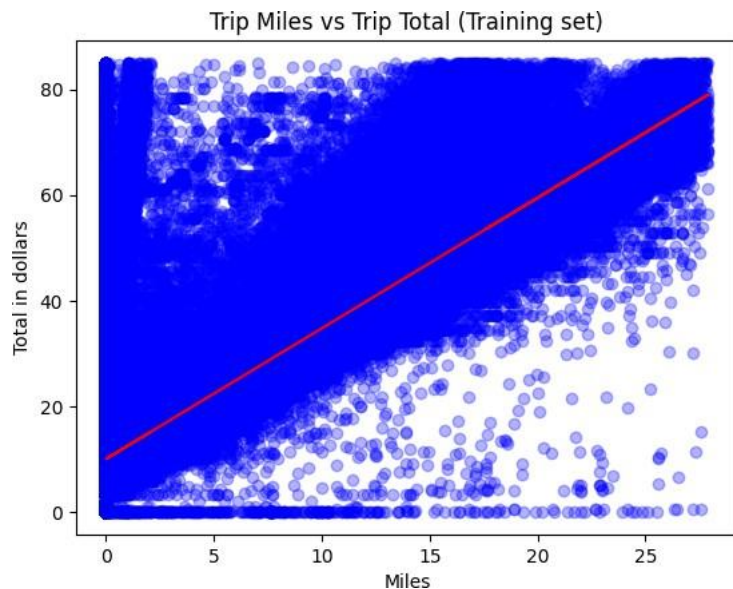
```
print("Coefficient:", model.coef_)
print("Intercept:", model.intercept_)
```

```
Coefficient: [[2.47145839]]  
Intercept: [10.02421236]
```

```
y_pred = model.predict(x_test)  
r2 = r2_score(y_test, y_pred)  
print("R2 Score:", r2)
```

R² Score: 0.6885674900078107

```
plt.scatter(x_train, y_train, color='blue', alpha=0.3)  
plt.plot(x_train, model.predict(x_train), color='red')  
plt.title('Trip Miles vs Trip Total (Training set)')  
plt.xlabel('Miles')  
plt.ylabel('Total in dollars')  
plt.show()
```



```
plt.scatter(x_test, y_test, color='blue', alpha=0.3)  
plt.plot(x_train, model.predict(x_train), color='red')  
plt.title('Trip Miles vs Trip Total (Testing set)')  
plt.xlabel('Miles')  
plt.ylabel('Total in dollars')  
plt.show()
```

